8 2026

dobachi

2026-08-12

- : 2026-08-01
- : 2026 7 28 16:27 M7.1 7 8
-

grounded-research
`check_spans.py`

- : `>` `-` `[S-xx]`
- :
- : §8

0.

2026 7 28 **7** 8 [S-01] 7 2016 10 3 3 [S-01] 36 179 8 1 11 30 [S-01] 465 9,872 [S-52] §1 §2

3

3

TSMC 7 [S-01] **5** [S-01] → §1.1 §3.1

[S-24] → §3.2 §3.3

[S-52] → §5

§10

§8

:

	§1
	§2
	§3
	§4
	§5
SNS	§6
IT	§11
	§8 §10

1.

Mj7.1 16km 7

Mj 7.1 16 km 7 — [S-01]

7 28 19 30 8 — [S-01]

10 3 3 7

28 2016 4 14 16 10 3 7 3 7 — [S-01]

2,438 Gal

2,438 Gal KiK-net 1,667 Gal — [S-01]

1.1 —

fab **5** 6

6 — [S-01]
5 — [S-01]

	7	—
	6	
	6	
	6	
	5	**JASM(TSMC)**

5

TSMC 5 1 — [S-42]

fab fab

1.2

— [S-01]

3 163

163 29 8 00 — [S-02]

2.

2.1 49 2026 8 11 16 00

49 8 11 16 00 **8 1 11 30** [S-56]

— [S-56] — [S-56]

8,611 330 8,603 327 [S-56]

39 26 327 353 392 1,133 1,045 8,603 10,781 — [S-56]

			8/1 11:30
	39	39	36
	26	27	6
	327	330	90
	1,133	1,133	179
	1,045	1,045	240
	8,603	8,611	1,021

—

47 8 9 10 00 [S-57]

— [S-57] 39 23 146 169 208 699 1,045 7,622 9,366 — [S-57]

8/1 11:30		36	6	90	179	240	1,021
8/9 10:00	47	**39**	23	146	699	**1,045**	7,622
8/11 16:00	49	39	26	**327**	**1,133**	1,045	8,603

- **8 9** 2
- **8 9** 1,045
- **8 11** 8/9→8/11 2 62% 2.2

10 **10**

2

§9

:

36 6 90 179 240 1,021 — [S-01] 36 6 93 179 240 1,029 — [S-01]

1 8 1 [S-56]

49 :

— [S-56] — [S-56] — [S-56]

36 1 2 39 [S-56] **49** **3**

8 1 34 36 **2** :

20 2 7 1 4 36 — [S-01]

2 49

— [S-56] — [S-56]

— [S-56]

2.2

:

12 7 — [S-01]

:

31 11 9 — [S-01]

:

28 3 — [S-01]

:

29 7 8 — [S-01]

2.3

84,000 **5 23**

84,000 — [S-03] 5 7/28 283 439 — [S-03]

4.9 PDF

4.9 31 ( ) — [S-04] `www.meti.go.jp` HTTP 403 CAPTCHA

Around 48,300 households in Kumamoto Prefecture lost power — [S-05]

2.4

29 12 465 9872 — [S-01]
11 **9**

17 30 8 — [S-01] 42 8 8 3,600 — [S-01]
11 9 — [S-01]

---

3. — fab

3.1

JASM TSMC

— [S-11] 2 29 — [S-11]
— 8 4 **2026-08-12** **2 7 31** 8 1 [S-59]
— **8 4** [S-58]

8 5 — [S-58] 3 — [S-58]

§3.4 3 **8 12**

8 4 8 — [S-06]
— 8 5

7 29 — [S-07] 8 5 3
— [S-07]

— [S-10]

JASM [S-11]

— [S-08] — [S-09]

— [S-12]

3.2 —

10 — [S-21] — [S-21]
7 31

3 8 5 — [S-16] 31 8 3 7 — [S-16] 31 2026
8 7 — [S-19] 32 7 —
[S-18] 8 5 2 — [S-20] — [S-18]

3.3 —

1 — [S-13]
— [S-13] — [S-13]
— [S-14]

3 110.8km

110.8km — [S-54] E3 ( )IC IC — [S-54] E3A
( )JCT ( )IC — [S-54] E77 JCT TB — [S-54]

3.4

but the impact on the global chip supply chain is expected to be limited, according to forecasts. — [S-23] TSMC's Kumamoto fab accounts for less than 3% of the company's total production capacity — [S-24] Analysts say the chip plants suffered reduced damage because seismic-resistant designs were put in place in advance, given Japan's frequent earthquakes. — [S-23]

— [S-26]
— [S-45] Even if wafer fabs avoid major damage, disruptions to transportation, power supply, or logistics could still result in short-term supply chain delays. — [S-24]

fab fab §3.2 3.3

---

4. fab

4.1

JASM 2021 11 9 **2016** **5 7**

70 8,000 *— [S-32] 2021 11 9 SSS 5 570*
20 — [S-32]

:

100 TSMC — [S-34]

PDF 403 :

— [S-31]

4.2 —

JASM §8

— [S-36] 30
— [S-35]

— [S-38]

— [S-36]

— [S-37]

4.3

V-Fine 6 100 200gal — [S-39]
— [S-39]

— [S-40] 0.6 600gal
— [S-40]

TSMC :

The clean rooms are isolated from external vibrations with double-shell structures, and there are viscous dampers and hydraulic pistons installed aplenty. — [S-41]

2,438 Gal 6 100 200gal

4.4

2011

3

— [S-48] 1 2,500 — [S-49]

6 → 6 — [S-49] 0.5 — [S-49]

2016 → **2026**

2016 3 — [S-50]

8 [S-06] §3.1 **3** → **2**

16 — [S-44]

TSMC

Taiwan experienced an earthquake on April 3 reaching magnitude level 7.2 (the largest in Taiwan over the past 25 years). The maximum magnitude measured at the Hsinchu, Longtan, and Zhunan science parks was 5, and reached magnitude 4 at the Taichung and Tainan science parks. — [S-51] overall tool recovery of our fabs reached more than 70% within 10 hours of the April 3 earthquake, with new fabs such as the Fab 18 facility reaching more than 80%. — [S-51]

5 5

4.5 —

7 28 1 — [S-44]
— [S-44]

2 [S-11] §3.1

— [S-22]

5.

28 2 64 — [S-52] minori 58 — [S-52]

465 9,872

---

6. SNS

8 SNS — [S-53]

7 29 10 30 7 30 [S-53]

AI 2024

---

7. Source Register

Tier T1= T2= T3=

ID	Source	URL	Tier	
S-01	(2026) - Wikipedia	https://ja.wikipedia.org/wiki/_%282026 %29	T3	
S-02	3 7/29 08	https://www.pref.kumamoto.jp/uploaded/attachment/315415.pdf	T1	
S-03	7/29 6:00	https://www.mlit.go.jp/common/002014126.pdf	T1	
S-04	8 8/1 15:30	https://www.meti.go.jp/2026_kumamoto/pdf/20260801_1.pdf	T1	403
S-05	2026 Kumamoto earthquake - Wikipedia (English)	https://en.wikipedia.org/wiki/2026_Kumamoto_earthquake	T3	
S-06	2 7/31	https://www.sony-semicon.com/ja/info/2026/2026073101.html	T2	
S-07	3	https://www.renesas.com/ja/about/newsroom/impact-26-earthquake-kumamoto-3	T2	

ID	Source	URL	Tier
S-08	7/28	https://www.tel.co.jp/news/disaster/2026/20260728_001.html	T2
S-09	2 7/29	https://www.tel.co.jp/news/disaster/2026/20260729_001.html	T2
S-10	EE Times Japan	https://ee-times.itmedia.co.jp/ee/articles/2607/28/news101.html	T3
S-11	ITmedia NEWS TSMC	https://www.itmedia.co.jp/news/article/2607/30/2000000305/	T3
S-12		https://kumanichi.com/articles/2019900	T3
S-13		https://kumanichi.com/articles/2020508	T3
S-14		https://www.nikkei.com/article/DGXZQOUC30AXO0Q6A730C2000000/	T3
S-16	3 7	https://www.nikkei.com/article/DGXZQOFD311MZ0R30C26A7000000/	T3
S-18		https://www.nikkei.com/article/DGXZQOUC312930R30C26A7000000/	T3
S-19	8 7	https://kumanichi.com/articles/2020503	T3

ID	Source	URL	Tier
S-20	8 5	https://carview.yahoo.co.jp/news/detail/3144bc08f34ad666d4d5f68b9ead5b79c4b64b5d/	T3
S-21		https://www.nikkei.com/article/DGXZQOFD29B3M0Z20C26A7000000/	T3
S-22		https://chemicaldaily.com/archives/835514	T3
S-23	Seoul Economic Daily Impact on Chip Supply Chain Seen Limited	https://en.sedaily.com/international/2026/07/29/kumamoto-quakes-impact-on-chip-supply-chain-seen-limited	T3
S-24	TrendForce 7.1 Kumamoto Earthquake: TSMC Confirms JASM Safe	https://www.trendforce.com/news/2026/07/29/news-7-1-kumamoto-earthquake-tsmc-confirms-jasm-safe-tel-halts-plants-as-chip-supply-chain-assesses-impact/	T3
S-26		https://www.nikkei.com/article/DGXZQOUC293G30Z20C26A7000000/	T3

ID	Source	URL	Tier	
S-31	JASM 2	https://www.meti.go.jp/policy/mono_info_service/joho/laws/semiconductor/semiconductor_plan/nintei_tokuteihandoutai_keikaku06.pdf	T1	403
S-32	2021 11 9	https://www.sony-semicon.com/ja/news/2021/2021110901.html	T2	
S-34		https://www.reri.or.jp/report-and-news/ckm-kumamoto-reasonwhy/	T3	
S-35	2022 11	https://www.cas.go.jp/jp/seisaku/keizai_anzen_hosyohousei/r4_dai4/siryou1.pdf	T1	
S-36	No221 2023.1	https://www.sangiin.go.jp/japanese/annai/chousa/keizai_prism/backnumber/r05pdf/202322101.pdf	T1	
S-37	6 2 9	https://www.mext.go.jp/content/20240209-mxt-jyohoka01-000033848_01.pdf	T1	

ID	Source	URL	Tier
S-38	5 6	https://www8.cao.go.jp/cstp/gaiyo/yusikisha/20230622/siryo2.pdf	T1
S-39		https://www.kajima.co.jp/tech/e_device/risk/tasoukai/index.html	T2
S-40		https://www.hirakata-g.co.jp/product/tackgel/	T2
S-41	Tom's Hardware TSMC factories rocked by magnitude 7.0 earthquake	https://www.tomshardware.com/tech-industry/semiconductors/tsmc-chipmaking-factories-rocked-by-magnitude-7-0-earthquake-that-was-the-strongest-in-27-years-but-facilities-escaped-unharmed-companys-earthquake-protection-measures-pay-off	T3
S-42	+IT	https://www.sbbit.jp/article/st/186349	T3

ID	Source	URL	Tier
S-44	16	https://news.yahoo.co.jp/articles/0d8c5473327ed649ff2dd27af98254c10db46b2f	T3
S-45		https://news.yahoo.co.jp/articles/d16da3f04e4e7cadb6dbb6e880cbe625826bcb19	T3
S-48	EE Times Japan	https://ee-times.itmedia.co.jp/ee/articles/1604/18/news117.html	T3
S-49		http://www.cocn.jp/material/ec9a4f2565e27e752d082827591d868e213ba413.pdf	T2
S-50	ITmedia 8	https://www.itmedia.co.jp/news/article/2607/31/2000000341/	T3
S-51	TSMC Updated Statement on April 3rd Earthquake SEC Form 6-K	https://www.sec.gov/Archives/edgar/data/1046179/000104617924000038/a20240405.htm	T1
S-52		https://kumanichi.com/articles/2019886	T3
S-53	piyolog 8 SNS	https://piy-olog.hatenadiary.jp/entry/2026/07/30/143946	T3
S-54	NEXCO 4 7/29 11:25	https://www.w-nexco.co.jp/emc/20260729112506.html	T1

ID	Source	URL	Tier
S-55	Open Data Spaces RAM IPA DADC / ODS-RAM V2	https://open-dataspaces.gitbook.io/ods-docs/jp/ods-ram/glossary.md	T1
S-56	49 R8.8.11 16:00	https://www.fdma.go.jp/disaster/info/items/20260728kumamoto-jishin49.pdf	T1
S-57	47 R8.8.9 10:00	https://www.fdma.go.jp/disaster/info/items/7595b20c947c985d45f6a34374b5f9ef31efbf94.pdf	T1
S-58	8 4 8/4	https://www.renesas.com/ja/about/newsroom/impact-26-earthquake-kumamoto-4	T2
S-59	2026-08-12 2	https://www.sony-semicon.com/ja/info/	T2

8.

JASM	TSMC	8	**NOT-FOUND**

	TSMC	**NOT-FOUND**
		NOT-FOUND
		NOT-FOUND 2
		NOT-FOUND

:

9.

- ~~34 36~~ → **2026-08-12** 49 1 2 §2.1
- 4.9 403
- PDF S-31 403
- 7/28
- **49 103,922 219,027 PDF**
- **2026-08-12** 47 49 4 **2 7/31** [S-59]
- : **8/4** JASM TSMC §3.3 8/1
- **2** §2.1
-
-
- §3.4

10.

2026-08-01 08-02 S-54 08-12 S-55 S-56

	107
FOUND	**105**

※ S-59 2 | NOT-FOUND | **0** | | FETCH-FAIL | 2 S-04 S-31 — 403 |

2026-08-12 17 S-55 4 S-56 9 S-57 2 S-58 2 S-56 S-57 PDF `pdftotext -layout`

2026-08-12

`citation_presence` **0 → 12** **12**

12 §X …

12 §0 §9 §10 §11

grounded-research :

```
python3 .../grounded-research/scripts/check_spans.py report.md
```

2026-08-02 : `check_spans.py` UTF-8 **Shift_JIS** **NOT-FOUND** S-10 EE Times Japan S-10 NOT-FOUND

3 NOT-FOUND … `[134]`

- span HTML: NFKC PDF: `pdftotext`
- : EE Times **Shift_JIS** UTF-8 NOT-FOUND ’ 6 SEC EDGAR **User-Agent** `www.meti.go.jp` `pr.tsmc.com` 403
- : SNS

11.

3

IDS / Gaia-X / Catena-X

11.1 —

3

#		
(a)		§3 fab §3.2 §3.3 —
(b)		§2.1 34 36
(c)		§6 AI

3 (a) (b) (c)

11.2

ODS-RAM Open Data Spaces Reference Architecture Model IPA DADC

Data Free Flow with Trust
— [S-55]

Open Data Spaces **ODS Protocols ODP** Open Dataspaces
— [S-55]

11.3 (a) —

§3.2 §3.4

11.4 (b) —

§2.1 34 36 §9

ODS-RAM

unified meta identifier UMI — [S-55]

2

ODS-RAM

Identity Layer: L3 — [S-55]

11.5 (c) —

§6 AI

`>` `-` `[S-xx]` §10

11.6

- 3
- 3
- (b)
-

11.7

- **ODS-RAM** 2026-08-12

  (c) ODS-RAM

- §11